

A COMPARISON OF CERTAIN TESTS FOR NORMALITY

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1. Introduction and Summary

Since a reasonably simple and generally acceptable alternative to normality has yet to be defined, optimal tests for normality cannot be constructed. Consequently tests for normality are constructed from very general considerations and no unique method of comparison of tests exists. Certain tests may be good for some alternatives whilst being rather poor for others.

In this paper we are primarily concerned with a test for normality based on small values of the statistic:

$$Q = \frac{\frac{1}{n} \sum_{i=1}^n \Phi^{-1}\left(\frac{i}{n+1}\right) X_{in}}{S}$$

where $X_{1n} < X_{2n} < \dots < X_{nn}$ are the order statistics of a random sample of size n , $\Phi(\cdot)$ is the cumulative distribution function of a $N(0,1)$ variate and S is the sample standard deviation. Since the distribution of Q is not known we cannot obtain the power of the test based on Q . In the special case where the population variance σ^2 is a known constant however, one may consider instead the test based on the statistic:

$$Q_n = \frac{\frac{1}{n} \sum_{i=1}^n \Phi^{-1}\left(\frac{i}{n+1}\right) X_{in}}{\sigma}$$

Intuitively one would expect a poor performance of the test based on Q_n to be indicative of a correspondingly poor performance of the test based on Q . The asymptotic distribution of Q_n is shown to be normal and the asymptotic power of the test based on Q_n is derived and compared with the power functions of the χ^2 -test, the Irwin-Kimball test and tests based on sample moments.

Using $\Delta(\Phi, F) = \inf_{\mu, \sigma} \sup_x |\Phi(x) - F(\mu + \sigma x)|$ as a measure of the departure of F from normality, it turns out that for the alternatives considered, the χ^2 -test is considerably more powerful than is suggested by the work of Kac, Kiefer and Wolfowitz (1955).

For both classes of alternatives considered the test based on Q_n is asymptotically less powerful than the χ^2 -test with a suitably chosen number of classes and tests based on sample moments whilst, in both cases, it is comparable with the Irwin-Kimball test.

A comparison of these results with the empirical results of Shapiro and Wilk (1965) for a test which is asymptotically equivalent to the Q_n -test for near

alternatives suggests that the expectation that the Q_n -test, would, asymptotically, be as good as the Q -test, is possibly groundless and that use of the known value of S in the manner indicated leads to an inferior test.

2. Motivation of the test statistics and the alternatives

Let $x_{1n} < x_{2n} < \dots < x_{nn}$ be the order statistics of a random sample of size n from a continuous distribution. The i -th order statistic of the sample is, asymptotically, likely to be relatively close to the percentile $\xi_{\frac{i}{n+1}}$ and if the sample

originates from a $N(\mu, \sigma^2)$ distribution it is to be expected that the points

$$(x_{in}, \Phi^{-1}(\frac{i}{n+1})), i = 1, 2, \dots, n$$

will be approximately on a straight line whilst if the sample does not come from a $N(\mu, \sigma^2)$ distribution, linearity is not to be expected. The degree of linearity between the $\{x_{in}\}$ and the $\{\Phi^{-1}(\frac{i}{n+1})\}$ therefore serves as an indication of normality. A measure of linearity of paired data is the ordinary product-moment correlation coefficient and consequently a test of normality is defined by the critical region:

$$\left\{ (x_{1n} < x_{2n} < \dots < x_{nn}) : \frac{\frac{1}{n} \sum_{i=1}^n \Phi^{-1}(\frac{i}{n+1}) x_{in} - \bar{x} \sum_{i=1}^n \Phi^{-1}(\frac{i}{n+1})}{\sqrt{S^2 \cdot \frac{1}{n} \sum_{i=1}^n \left\{ \Phi^{-1}(\frac{i}{n+1}) - \frac{1}{n} \sum_{i=1}^n \Phi^{-1}(\frac{i}{n+1}) \right\}^2}} < C_0 < 1 \right\}$$

Since $\sum_{i=1}^n \Phi^{-1}(\frac{i}{n+1}) = 0$ and $\frac{1}{n} \sum_{i=1}^n \left\{ \Phi^{-1}(\frac{i}{n+1}) \right\}^2$ is, for every n , a known constant which tends to 1 from the left, the test is effectively:

$$\left\{ (x_{1n} < x_{2n} < \dots < x_{nn}) : \frac{\frac{1}{n} \sum_{i=1}^n \Phi^{-1}(\frac{i}{n+1}) x_{in}}{S} < C \right\}$$

The statistic:

$$Q = \frac{\frac{1}{n} \sum_{i=1}^n \Phi^{-1}(\frac{i}{n+1}) x_{in}}{S}$$

is a special case of a general class of statistics

$$\frac{\frac{1}{n} \sum_{i=1}^n \epsilon_{in} X_{in}}{S}$$

where $\epsilon_{1n}, \dots, \epsilon_{nn}$ are constants. A special case of this class was considered em-

pirically by Shapiro and Wilk (1965), their W_n -test statistic being asymptotically equivalent to Q^2 for near alternatives. In unpublished work Venter (1967) considered this class of statistics and derived the asymptotic distribution for a particular choice of the ϵ_{in} . The distribution and the properties of the test based on Q are not known; the empirical results of Shapiro and Wilk and other results of Venter (1968) suggest however that the limiting distribution of Q under the null hypothesis may not be normal. One of the main difficulties concerning the distribution of Q in the case of normality is that the asymptotic distribution of $\sqrt{n}(Q-S)$ is degenerate.

If we let:

$$Q_n = \frac{\frac{1}{n} \sum_{i=1}^n \Phi^{-1}\left(\frac{i}{n+1}\right) X_{in}}{\sigma}$$

where σ^2 is the known variance of the underlying distribution, the asymptotic distribution of Q_n can be derived for a large class of underlying distributions and hence it is easier to investigate the properties of the test based on Q_n than those of the test based on Q . We therefore restrict our attention to the case where the hypothesis to be tested is the simple null hypothesis that the sample originates from a $N(0,1)$ distribution.

The well-known χ^2 -test needs no further motivation here. One need only point out that if the number of classes is allowed to tend to infinity with sample size the test can, as yet, only be used when the null hypothesis is a simple hypothesis. To be specific we will assume that the χ^2 -test is carried out by using $[n^k]$ classes where $0 < k \leq 1$ and where the classes are equi-probable under the null-hypothesis.

The Irwin-Kimball test can, as yet, only be used when the null hypothesis is a simple hypothesis. When specialised to the case where the null hypothesis is that the sample originates from a $N(0,1)$ distribution the test statistic is

$$V(n) = n \sum_{i=1}^{n+1} \left\{ \Phi(X_{in}) - \Phi(X_{i-1,n}) \right\}^2$$

where we define $X_{0n} = -\infty$ and $X_{n+1,n} = \infty$. Basically this test compares the distances

$$\left\{ \Phi(X_{in}) - \Phi(X_{i-1,n}) \right\}$$

with the expected distances $\frac{1}{n+1}$ between the successive order statistics of the uniform distribution and rejects the null hypothesis if the sum of the squared "errors" is too large.

The three tests mentioned above are general tests for normality, their common features being that they have power for all non-normal alternatives with $\mu = 0$ and $\sigma^2 = 1$ and for such alternatives they are all one-tailed tests. In contrast tests for normality based on sample moments cannot justifiably be called tests for normality. In general such tests are two-tailed tests and we will compare them with the other tests because of their historical and practical importance. In particular we will, for purpose of comparison, consider tests based on $m'_3 =$

$$\frac{1}{n} \sum X_{in}^3 \text{ and on } m_4^* = \frac{1}{n} \sum X_{in}^4.$$

Before tests for normality can be compared, suitable alternatives have to be formulated. Kac, Kiefer and Wolfowitz (1955) considered as alternatives the class of distributions F such that $\Delta(\Phi, F) = O(n^{-p})$ and for each test involved in their comparison, they found the greatest value of p for which the test is asymptotically unbiased. The greater p , the "better" or the more "sensitive" the test. This class of alternatives is very large, containing, inter alia, all possible pathological cases which can be constructed to reduce the power of the test and all alternatives which are irrelevant to the situation in which the test is to be used.

Occasionally one has some knowledge, albeit scant, of the type of alternative one requires the test to be sensitive to, and a case can be made for comparing tests of normality for classes of alternatives less inclusive than the class considered by Kac, Kiefer and Wolfowitz. Likewise a case can be made for constructing tests for normality which are intended to have high power for certain types of departure from normality.

The null hypothesis is:

$$H_0 : F(x) = \Phi(x)$$

and as alternatives we will consider separately

(i) the class H of distribution functions $F(., \theta)$ such that $F(., \theta)$ satisfies certain regularity conditions to be specified later and such that:

$$F(x; \theta) = \Phi(x) + \Phi'(x) \sum_{j=2}^{\infty} a_j \theta^{j+1} H_j(x)$$

where $H_j(.)$ is the j -th Hermite polynomial;

(ii) the class L of distributions $F(x; \theta)$ such that

$$F(x; \theta) = (1 - \theta)\Phi(x) + \theta G(x)$$

where the standardized distribution function $G(.)$ with uniformly continuous density function $G'(.)$ is supported on the interval $[a, b]$. For both classes of alternatives we will let $\theta = n^{-r}$ where $r > 0$ and n is the sample size.

If, for some alternative $F(x; \theta)$ with $\theta = n^{-r}$:

$$\Delta(\delta, F) = \inf_{\mu, \sigma} \sup_x |\Phi(x) - F(\mu + \sigma x; \theta)| \geq A n^{-p}$$

where $0 < A < \infty$ we will say that F belongs to the class $K(p)$. Our intention is to find for each test the greatest value of p for which the test is asymptotically unbiased for an alternative F of a specified type in $K(p)$.

3. The Asymptotic Power of the Q_n -Test

Theorem 1

Let $X_{1n} < X_{2n} < \dots < X_{nn}$ be the order statistics of a random sample of size n from a continuous distribution with continuous distribution function $F(., \theta)$ such that:

(i) $F^{-1}(\cdot, \theta)$ is continuous on $(0, 1)$ and satisfies a first-order Lipschitz condition in every interval bounded away from 0 and 1.

(ii) $\frac{dF^{-1}(x; \theta)}{dx}$ is continuous for $x \in (0, 1)$ with the possible exception of a finite number of points in $(0, 1)$. [Define $\frac{dF^{-1}(x; \theta)}{dx} = 0$ if it is not otherwise defined.]

(iii) (tail smoothness) there exists a $\delta_0, 0 < \delta_0 < 1$ such that for each $K > 0$ and $0 < u_1, u_2 < \delta_0$ for which $K^{-1} < u_1/u_2 < K$ there exists a finite M such that

$$M^{-1} < \frac{dF^{-1}(u_1; \theta)}{du_1} / \frac{dF^{-1}(u_2; \theta)}{du_2} < M$$

and, for each $K > 0$ and $1 - \delta_0 < u_1, u_2 < 1$ such that $K^{-1} < 1 - u_1 / 1 - u_2 < K$ there exists a finite M such that

$$M^{-1} < \frac{dF^{-1}(u_1; \theta)}{du_1} / \frac{dF^{-1}(u_2; \theta)}{du_2} < M$$

(iv) The integrals

$$B = \int_0^1 \{x(1-x)\}^{1/2} \Phi^1(x) \frac{dF^{-1}(x; \theta)}{dx} dx$$

$$\sigma^2(\theta) = \frac{2}{\sigma^2} \int_0^1 \int_0^1 x(1-y) \Phi^1(x) \frac{dF^{-1}(x; \theta)}{dx} \Phi^1(y) \frac{dF^{-1}(y; \theta)}{dy} dx dy$$

converge absolutely and $\sigma^2(\theta) > 0$.

$$(v) \left| \frac{1}{n} \sum_{j=1}^n \Phi^1\left(\frac{j}{n+1}\right) F^{-1}\left(\frac{j}{n+1}; \theta\right) - \int_0^1 \Phi^1(x) F^{-1}(x; \theta) dx \right| = o(n^{-1/2})$$

$$\text{and } \mu(\theta) = \int_0^1 \Phi^1(x) F^{-1}(x; \theta) dx$$

then

$$\mathcal{L}\left\{\frac{\sqrt{n}[Q_n - \mu(\theta)]}{\sigma(\theta)}\right\} \xrightarrow[n \rightarrow \infty]{} N(0, 1)$$

Proof

In the form stated the theorem is simply a special case of the more general theorem 3 of Chernoff, Gastwirth and Johns (1967).

Theorem 2

If $F(x; \theta) = \Phi(x)$ then $\mathcal{L}\left\{\sqrt{2n}(Q_n - 1)\right\} \xrightarrow[n \rightarrow \infty]{} N(0, 1)$.

Proof

Although the author has not been able to find a complete proof in the literature, the theorem seems to be well known (see e.g. example 3 on p. 70 of Chernoff, Gastwirth and Johns (1967)). The proof, which follows readily from theorem 1, will therefore not be given here.

Theorem 1 does not make provision for the dependence of F on n . If however the theorem is valid uniformly in θ for $\theta \in [0, \delta]$ one may substitute n^{-1} for θ . It can be seen that this substitution may be made by noting that if $F(\cdot, \theta)$ satisfies

the hypotheses of theorem 1 uniformly in θ for $\theta \in [0, \delta]$ the conclusion of theorem 1 applies uniformly in θ . If therefore $\psi(n, \theta, a)$ is the m.g.f. of

$$\frac{Q_n - \mu(\theta)}{\sigma(\theta)}$$

then, given an arbitrary $\epsilon > 0$, there exists a finite $N(\delta)$ such that $|\psi(n, \theta, a) - e^{1/2 a^2}| < \epsilon$ for $n > N(\delta)$ and all $\theta \in [0, \delta]$, in particular also for $\theta = n^{-r}$ where $r > 0$.

We now define H to be the class of distribution functions $F(x; \theta)$ for which:

$$F(x; \theta) = \Phi(x) + \Phi'(x) \sum_{j=2}^{\infty} a_j \theta^{j+1} H_j(x)$$

where $F(x; \theta)$ satisfies the hypotheses of theorem 1 uniformly in θ for $\theta \in [0, \delta]$ in such a way that

$$\sigma^2(\theta) = 1/2 + o(1) \text{ if } \theta = n^{-r}.$$

Theorem 3

The asymptotic power of the Q_n test for the classes of alternatives H and L is:

$$1 - \Phi(Z_\alpha - \lim_{n \rightarrow \infty} \sqrt{2n} \{1 - \mu(\theta)\})$$

where $\Phi(Z_\alpha) = 1 - \alpha$ and α is the size of the test, and

$$\mu(\theta) = \int_0^1 \Phi \Psi(x) F^{-1}(x; \theta) dx$$

Proof

For the class of alternatives H the theorem follows immediately from theorem 1 and the definition of the class H .

For the class of alternatives L we note that conditions (i) and (ii) of theorem 1 are satisfied for all $\theta \in [0, \delta]$ where we take $\delta < 1$. For condition (iii) of theorem 1 we choose for the left-hand tail of $F(x; \theta)$: $\delta_0 < (1 - \delta) \Phi(a)$; then if $0 < u_1, u_2 < \delta_0$ we have:

$$\frac{dF^{-1}(u_1; \theta)}{du_1} / \frac{dF^{-1}(u_2; \theta)}{du_2} = \frac{\Phi' \left\{ \Phi^{-1} \left(\frac{u_2}{1 - \theta} \right) \right\}}{\Phi' \left\{ \Phi^{-1} \left(\frac{u_1}{1 - \theta} \right) \right\}}$$

and from the proof of theorem 2 it follows that for a given $K > 0$ such that $K^{-1} < u_1/u_2 < K$ it is possible to find a finite $M(\delta)$ such that

$$\{M(\delta)\}^{-1} < \frac{\Phi' \left\{ \Phi^{-1} \left(\frac{u_2}{1 - \theta} \right) \right\}}{\Phi' \left\{ \Phi^{-1} \left(\frac{u_1}{1 - \theta} \right) \right\}} < M(\delta)$$

for all $\theta \in [0, \delta]$. A similar result applies for the right-hand tail of $F(x; \theta)$. For condition (iv) we have:

$$\int_0^1 \{x(1-x)\}^{1/2} \Phi^{-1}(x) \frac{dF^{-1}(x; \theta)}{dx} dx = \int_{-\infty}^{\infty} [\Phi(x)\{1 - \Phi(x)\}]^{1/2} \frac{\Phi'(x) dx}{F^{-1}[\Phi^{-1}\{\Phi(x)\}; \theta]}$$

where:

$$0 < \frac{\Phi'(x)}{F'[F^{-1}\{\Phi(x)\}; \theta]} \left\{ \sqrt{2\pi} \inf_{a \leq x \leq b} \Phi'(x) \right\}^{-1} (1-\theta)^{-1} < \infty$$

for

$$x \in [F^{-1}\{(1-\theta)\Phi(a)\}, F^{-1}\{\theta + (1-\theta)\Phi(b)\}] \quad \text{and}$$

$$0 < \frac{\Phi'(x)}{F'[F^{-1}\{\Phi(x)\}; \theta]} < \frac{1}{1-\theta} < \infty$$

otherwise. From this and the proof of theorem 2 it follows that the above integral for $\sigma(\theta)$ converges absolutely. The proof that

$$\left| \int_0^1 \Phi^{-1}(x) F^{-1}(x; \theta) dx - \frac{1}{n} \sum_{j=1}^n \Phi^{-1}\left(\frac{j}{n+1}\right) F^{-1}\left(\frac{j}{n+1}; \theta\right) \right| = o(n^{-1/2})$$

uniformly in θ for $\theta \in (0, \delta)$ follows trivially from the proof of theorem 2 and the fact that $\Phi^{-1}(x)F^{-1}(x; \theta)$ is well behaved for all $x \in [(1-\theta)\Phi(a), \theta + (1-\theta)\Phi(b)]$ and all $\theta \in [0, \delta]$. Finally $\sigma^2(\theta) \rightarrow 1$ as $\theta \rightarrow 0$ for the class L as can be seen by taking the limit under the defining integrals.

For the class of alternatives H we find:

$$\mu(\theta) = 1 - a_2^2 \theta^6 - 3a_3^2 \theta^8 - 12a_4^2 \theta^{10} - 24a_2^2 a_3 \theta^{10} + O(\theta^{11})$$

and from theorem 3 it follows on putting $\theta = n^{-1}$ that the Q_n -test is asymptotically unbiased for all alternatives of the class H which are in $K(\frac{1}{4})$. For the class of alternatives L:

$$\begin{aligned} \mu(\theta) &= 1 + \theta \int_{-\infty}^{\infty} x \{G(x) - \Phi(x)\} dx \\ &+ \theta^2 \int_{-\infty}^{\infty} x [G(x) - \Phi(x)] \left\{ \frac{1}{2} x [G(x) - \Phi(x)] + G'(x) - \Phi'(x) \right\} \Phi'(x) dx + O(\theta^3) \end{aligned}$$

For any standardized distribution function $G(\cdot)$ the first integral on the right is zero and the Q_n -test will only be asymptotically unbiased for alternatives of the class L in $K(\frac{1}{4})$ if:

$$\int_{-\infty}^{\infty} x [G(x) - \Phi(x)] \left\{ \frac{1}{2} x [G(x) - \Phi(x)] + G'(x) \right\} \Phi'(x) dx \neq 0, \text{ i.e. if}$$

$$\int_{-\infty}^{\infty} \frac{[G(x) - \Phi(x)]^2}{\Phi'(x)} dx \neq 0.$$

The Q_n -test is therefore asymptotically unbiased for all alternatives of the class L in $K(\frac{1}{4})$.

It only remains to show that the class of alternatives H is not empty. To this end we may consider the following three alternative distribution functions:

$$H_{a_1}: F(x; \theta) = (1-p) \Phi\left(\frac{x}{\sqrt{1 - \frac{p\theta^2}{1-p}}}\right) + p \Phi\left(\frac{x}{\sqrt{1 + \theta^2}}\right)$$

$$H_{a_2}: F(x; \theta) = (1-p) \Phi\left(\frac{x + \frac{p\theta}{1-p}}{\sqrt{1 - \frac{p\theta^2}{1-p}}}\right) + p \Phi\left(\frac{x - \theta}{\sqrt{1 - \frac{p\theta^2}{1-p}}}\right)$$

$$H_{a_3}: F(x; \theta) = (1-p) \Phi\left(x + \frac{p\theta}{1-p}\right) + p \Phi\left(\frac{x - \theta}{\sqrt{1 - \frac{\theta^2}{1-p}}}\right)$$

The distribution defined by H_{a_1} is a mixture of a $N(0, 1 - \frac{p\theta^2}{1-p})$ and a $N(0, 1 + \theta^2)$ distribution, the distribution defined by H_{a_2} is a mixture of a $N(-\frac{p\theta}{1-p}, 1 - \frac{p\theta^2}{1-p})$ and a $N(\theta, 1 - \frac{p\theta^2}{1-p})$ distribution whilst that defined by H_{a_3} is a mixture of a $N(\frac{p\theta}{1-p}, 1)$ and a $N(\theta, 1 - \frac{\theta^2}{1-p})$ distribution.

These three alternatives represent three different ways in which an alternative can be written as the mixture of two normal distributions subject to the restrictions imposed above. In each case the parameters have been inserted in such a way that the mean and variance of the alternative distribution are 0 and 1 respectively. For each of these three alternatives:

$$\int_{-\infty}^{\infty} \frac{dF(x; \theta)}{dx} \frac{x^2}{e^4} dx \text{ exists and } \frac{dF(x; \theta)}{dx}$$

is of bounded variation in $(-\infty, \infty)$. It follows therefore (cf. Cramer (1946)) that in each of these cases we can write $F(., \theta)$ and $F'(., \theta)$ as the convergent series:

$$F(x; \theta) = \Phi(x) + \Phi'(x) \sum_{j=2}^{\infty} a_j(p) \theta^{j+1} H_j(x)$$

$$F'(x; \theta) = \Phi'(x) \left\{ 1 - \sum_{j=2}^{\infty} a_j(p) \theta^{j+1} H_{j+1}(x) \right\}$$

where some of the $a_j(p)$ may be zero. Since the proof that each of the above three distributions satisfies the conditions of theorem 1 and the other imposed condition is roughly the same in all three cases we only consider the proof for H_{a_1} here in detail.

Given $F(x; \theta) = (1-p) \Phi\left(\frac{x}{\sqrt{1 - \frac{p\theta^2}{1-p}}}\right) + p \Phi\left(\frac{x}{\sqrt{1 + \theta^2}}\right)$ it is clear that $F(., \theta)$ and

$F^{-1}(., \theta)$ are continuous and have continuous derivatives and that $F^{-1}(., \theta)$ satisfies the necessary first-order Lipschitz condition uniformly in θ . We must now show that there exists a fixed θ -interval including $\theta = 0$ for which:

$$\int_0^1 [x(1-x)]^{1/2} \Phi^{-1}(x) \frac{dF^{-1}(x;\theta)}{dx} \quad \text{and}$$

$$\int_0^1 \int_0^1 x(1-y) \Phi^{-1}(x) \frac{dF^{-1}(x;\theta)}{dx} \Phi^{-1}(y) \frac{dF^{-1}(y;\theta)}{dy} dx dy$$

converge absolutely uniformly in θ . Since the two integrals above converge absolutely when $F(x; \theta) = \Phi(x)$ and since we now have:

$$\frac{dF^{-1}(x;\theta)}{dx} = \left\{ \frac{1-p}{\sigma_1} \Phi' \left[\frac{F^{-1}(x;\theta)}{\sigma_1} \right] + \frac{p}{\sigma_2} \Phi' \left[\frac{F^{-1}(x;\theta)}{\sigma_2} \right] \right\}^{-1}$$

where $\sigma_1^2 = 1 - \frac{p\theta^2}{1-p}$ and $\sigma_2^2 = 1 + \theta^2$; it is sufficient to show that:

$$\Phi'[\Phi^{-1}(x)] \left\{ \frac{1-p}{\sigma_1} \Phi' \left[\frac{F^{-1}(x;\theta)}{\sigma_1} \right] + \frac{p}{\sigma_2} \Phi' \left[\frac{F^{-1}(x;\theta)}{\sigma_2} \right] \right\}^{-1} \geq 0$$

is bounded for all $x \in (0,1)$ and all δ in some fixed θ -interval $[0, \delta]$. Since

$$\sigma_1 |\Phi^{-1}(x)| \leq |F^{-1}(x;\theta)| \leq \sigma_2 |\Phi^{-1}(x)|$$

for all $x \in (0,1)$ we have:

$$\{F^{-1}(x;\theta)\}^2 / \sigma_2^2 - \{\Phi^{-1}(x)\}^2 > 0 \quad \text{so that:}$$

$$\Phi'[\Phi^{-1}(x)] / \Phi' \left[\frac{F^{-1}(x;\theta)}{\sigma_2} \right] \geq 1 \quad \text{and} \quad \Phi'[\Phi^{-1}(x)] / \Phi' \left[\frac{F^{-1}(x;\theta)}{\sigma_1} \right] \leq \frac{\sigma_2}{p}$$

It follows therefore that the two integrals converge absolutely uniformly in θ for $\theta \in [0, \delta]$ where e.g. $\delta^2 < (1-p)/p$.

As for tail smoothness we must show that there exist finite bounds M and M^{-1} on

$$\frac{dF^{-1}(u_1;\theta)}{du_1} / \frac{dF^{-1}(u_2;\theta)}{du_2}$$

in the upper and lower tails of $F(\cdot, \theta)$ which depend only on K and δ where $K^{-1} < u_1/u_2 < K$ and $\theta \in [0, \delta]$. Such bounds do exist for the three alternatives given. Specifically, in the case of H_{a1} the situation for the two tails is similar and we consider only the lower tail here. For this it must, in effect, be shown that if $0 < c < \infty$ then for $0 < u_2 < \delta_0 < 1$:

$$M^{-1} < \frac{\beta_1 \Phi' \left[\frac{F^{-1}(u_2;\theta)}{\sigma_1} \right] + \beta_2 \Phi' \left[\frac{F^{-1}(u_2;\theta)}{\sigma_2} \right]}{\beta_1 \Phi' \left[\frac{F^{-1}(cu_2;\theta)}{\sigma_1} \right] + \beta_2 \Phi' \left[\frac{F^{-1}(cu_2;\theta)}{\sigma_2} \right]} < M$$

for all values of $\theta \in [0, \delta]$ where β_1 and β_2 are constants and σ_1 and σ_2 are as defined above. If $\delta^2 < \frac{1-p}{p}$ the fraction in question is continuous on the interval

$[0, \delta_0]$ uniformly in θ and hence it is sufficient to show that the value of the fraction tends to a finite limit as $u_2 \rightarrow 0$; i.e. the discontinuity at the origin is removable. Applying L' Hospital's rule once it can be seen that it is sufficient to show that $F^{-1}(u; \theta)/F^{-1}(cu; \theta)$ tends to a finite limit as u tends to zero for any finite c . We have:

$$\lim_{u \rightarrow 0} \frac{F^{-1}(u; \theta)}{F^{-1}(cu; \theta)} \leq \lim_{u \rightarrow 0} \frac{\sigma_2 \Phi^{-1}(u)}{\sigma_1 \Phi^{-1}(cu)} = \frac{\sigma_2}{\sigma_1}$$

so that the tail smoothness condition is satisfied for the lower tail uniformly in θ for $\theta \in [0, \sqrt{\frac{1-p}{p}}]$.

If $\theta = O(n^{-r})$ where $r > 0$, then:

$$\sigma_1^2 = 2 \int_0^1 \int_0^1 x(1-y) \Phi^{-1}(x) \frac{dF^{-1}(x; \theta)}{dx} \Phi^{-1}(y) \frac{dF^{-1}(y; \theta)}{dy} dx dy = 1/2 + O(\theta^2)$$

whilst:

$$\mu(\theta) = \int_0^1 \Phi^{-1}(x) F^{-1}(x; \theta) dx = 1 - \frac{3p^2\theta^8}{64(1-p)^2} + O(\theta^{10})$$

and

$$|\mu(\theta) - \frac{1}{n} \sum_{i=1}^n \Phi^{-1}\left(\frac{j}{n+1}\right) F^{-1}\left(\frac{j}{n+1}\right)| < o(n^{-1/2})$$

4. The asymptotic power of the χ^2 -test

If we test for normality by using Pearson's χ^2 with $[n^k]$ classes where n is the sample size, $[n^k]$ = largest integer less than or equal to n^k , $0 < k \leq 1$, and where the classes are equi-probable if H_0 is true, the i -th class is the interval

$$\left[\Phi^{-1}\left(\frac{i-1}{[n^k]}\right), \Phi^{-1}\left(\frac{i}{[n^k]}\right) \right] \text{ for } i = 1, 2, \dots, [n^k]$$

and we let p_i denote the probability of this interval if H_a is true. For large n the test statistic is:

$$\chi_n^2 = \sum_{i=1}^{[n^k]} \frac{(N_i - n^{1-k})^2}{n^{1-k}}$$

where N_i is the frequency of the i -th class. It follows from a result of Steck (1957) that for $0 < k \leq 1$:

$$\mathcal{L} \left\{ \frac{\chi_n^2 - n^k}{\sqrt{2n^k}} \mid H_0 \right\} \xrightarrow[n \rightarrow \infty]{} N(0, 1).$$

We have identically:

$$\frac{\chi_n^2 - n^k - \sum_{i=1}^{[n^k]} (1 - n^k p_i)^2}{\sqrt{2n^k}} = \frac{\sum_{i=1}^{[n^k]} \frac{(N_i - np_i)^2}{np_i} - n^k - \sum_{i=1}^{[n^k]} \left(\frac{N_i - np_i}{\sqrt{np_i}} \right)^2 (1 - n^k p_i) - 2 \sum_{i=1}^{[n^k]} (N_i - np_i)(1 - n^k p_i)}{\sqrt{2n^k}}$$

If $n^k p_i = 1 + o(1)$ for all p_i and if

$$n^{1-k} \sum_{i=1}^{[n^k]} (1 - n^k p_i)^2 = O(n^{\frac{k}{2}})$$

it can be shown that:

$$\sum_{i=1}^{[n^k]} \left(\frac{N_i - np_i}{\sqrt{np_i}} \right)^2 (1 - n^k p_i) = o_p(n^{\frac{k}{2}}) \text{ and}$$

$$\sum_{i=1}^{[n^k]} (N_i - np_i)(1 - n^k p_i) = o_p(n^{\frac{k}{2}}) \text{ so that:}$$

$$\mathcal{L} \left\{ \frac{\chi_n^2 - n^k - n^{1-k} \sum_{i=1}^{[n^k]} (1 - n^k p_i)^2}{\sqrt{2n^k}} \mid H_a \right\} \xrightarrow{n} N(0, 1)$$

Clearly $n^k p_i = 1 + o(1)$ for both classes of alternatives and it only remains to select $\theta = n^{-r}$ and k in such a way that

$$n^{1-k} \sum_{i=1}^{n^k} (1 - n^k p_i)^2 = O(n^{\frac{k}{2}})$$

for asymptotic normality to hold. For a suitable choice of r and k we also have:

$$n^{1-k} \sum_{i=1}^{[n^k]} (1 - n^k p_i)^2 \xrightarrow{n} n \int_0^1 \left(1 - \frac{dF[\Phi^{-1}(x); \theta]}{dx} \right)^2 dx$$

and, on making this choice, evaluating the integral and retaining only the significant terms we obtain the following powers for the χ^2 -test:

For the class L:

$$P \left[\frac{\chi^2 - n^k}{\sqrt{2n^k}} > Z_\alpha \mid H_a \right] = \begin{cases} Z_\alpha & \text{for } 1-2r < \frac{k}{2} \\ 1 - \Phi \left(Z_\alpha - \frac{1}{\sqrt{2}} \right) & \text{for } 1-2r = \frac{k}{2} \\ 1 & \text{for } 1-2r > \frac{k}{2} \end{cases}$$

$$\text{where } I = \int_a^b \left(\frac{G'(x)}{\Phi'(x)} \right)^2 d\Phi(x) - 1 > \sqrt{\frac{\pi}{2}} - 1 > 0$$

For the class H:

$$P \left[\frac{\chi^2 - n^k}{\sqrt{2n^k}} > Z_\alpha \mid H_a \right] = \begin{cases} Z_\alpha & \text{for } 1-2(s+1)r < \frac{k}{2} \\ 1 - \Phi \left(Z_\alpha - \frac{1}{\sqrt{2}} \right) & \text{for } 1-2(s+1)r = \frac{k}{2} \\ 1 & \text{for } 1-2(s+1)r > \frac{k}{2} \end{cases}$$

where $I = a_s^2(s+1)!$ and s is the smallest integer for which $a_j \neq 0$ in the series

$$F(x; \theta) = \Phi(x) + \Phi'(x) \sum_{j=2}^{\infty} a_j \theta^{j+1} H_j(x).$$

It follows immediately therefore that for both alternatives the number of classes can be chosen in such a way that the χ^2 -test is asymptotically unbiased for alternatives $F \in K(1/2, \delta)$ for any $\delta > 0$.

The reason for the increased sensitivity of the χ^2 -test is, in effect, that the assumption that

$$n^k > \left\{ \sup_x |\Phi(x) - F(x)| \right\}^{-1}$$

made on page 313 of Mann and Wald (1942) is no longer necessary. This condition is required for the class of all alternatives to guard against the pathological case where the null and alternative distributions coincide excepting on a set of measure n^{-k} . [By coincide we don't mean $F(x) = \Phi(x)$ but $F(\Lambda) = \Phi(\Lambda)$ for any Borel set Λ .] In view of this it does not seem justifiable to advocate, as on page 439 of Kendall and Stuart (1961), that one should increase the number of classes in proportion to $n^{2/5}$. For "smooth" alternatives it appears to be advantageous to increase the number of classes at a much slower rate.

The asymptotic relative efficiency of the χ^2 test with n classes relative to the Q_n -test for the class of alternatives H is 900% for an alternative with skewness ($a_2 \neq 0$) and 1600% for one with no skewness but having a departure in kurtosis ($a_2 = 0, a_3 \neq 0$).

5. The asymptotic power of the Irwin-Kimball test

Let $X_{1n} \leq X_{2n} \leq \dots \leq X_{nn}$ be the order statistics for a random sample of size n from the distribution of a continuous random variable X and suppose the statistic

$$V(n) = n \sum_{i=1}^{n+1} \{ \Phi(X_{in}) - \Phi(X_{i-1n}) \}^2$$

with $\Phi(X_{n+1n}) = 1$ and $\Phi(X_{0n}) = 0$ is used to test the hypothesis that the distribution function of X is $\Phi(\cdot)$. As shown by Weiss (1965) the power of this test is

$$1 - \Phi(Z_{\alpha} - n^{1/2-2\delta} \int_0^1 r^2(y) dy)$$

if the density of $\Phi(X)$ can be written in the form $f_n(x) = 1 + n^{-\delta} r(x)$ for $0 < x < 1$ where $\delta > 0, \int_0^1 r(x) dx = 0$ and $|r''(x)| < D < \infty$ for $0 \leq x \leq 1$. For the class of alternatives L the density of $\Phi(X)$ can be written in the required form and can be shown to satisfy the conditions of Weiss' result so that, for this class of alternatives, the power of the test is, with $\theta = n^{-r}$:

$$1 - \Phi(Z_{\alpha} - n^{1/2-2r} \int_{-\infty}^{\infty} \left\{ 1 - \frac{G'(x)}{\Phi'(x)} \right\}^2 d\Phi(x))$$

The test is therefore asymptotically unbiased for alternatives of the given type in $K(\frac{1}{4})$.

For the class H the density of $\Phi(X)$ can be written in the form

$$f_n(x) = 1 + n^{-\delta} r_n(x)$$

so that Weiss' result cannot be applied because of the dependence of $r_n(\cdot)$ on n . Although Weiss' result can be improved to allow for such dependence the regularity conditions imposed on the second derivative of $\Phi(X)$ are too stringent for all alternatives in the class H. It is possible however to calculate the asymptotic mean and variance of the test statistic for an alternative in the class H and, using these and the well-known Chebyshev inequality it can be shown that the Irwin-Kimball test can at most be asymptotically unbiased for alternatives of the class H in $K(\frac{1}{4})$. For both classes of alternatives therefore, the test is comparable with the Q_n -test and with the χ^2 -test with n equi-probable classes under H_0 .

6. The asymptotic power of tests based on m'_3 and m'_4

A straight-forward application of the central limit theorem yields the following asymptotic powers:

$$\text{For } m'_3: 1 - \Phi\left(\frac{Z_\alpha}{2} \frac{\sqrt{n}\mu'_3}{\sqrt{15}}\right) + \Phi\left(-\frac{Z_\alpha}{2} \frac{\sqrt{n}\mu'_3}{\sqrt{15}}\right)$$

$$\text{For } m'_4: 1 - \Phi\left(\frac{Z_\alpha}{2} + \sqrt{\frac{n}{96}} \{3 - \mu'_4\}\right) + \Phi\left(-\frac{Z_\alpha}{2} + \sqrt{\frac{n}{96}} \{3 - \mu'_4\}\right)$$

For both classes of alternatives the test based on m'_3 will be asymptotically unbiased for alternatives in $K(\frac{1}{2})$ if $\mu'_3 = 0$. If $\mu'_3 = 0$ and $\mu'_4 = 3$ however, the test based on m'_4 is asymptotically unbiased for alternatives in $K(\frac{3}{8})$ in the case of the class H and for alternatives in $K(\frac{1}{2})$ in the case of the class L. In both cases these tests have no power if $\mu'_3 = 0$ and $\mu'_4 = 3$.

7. Discussion

The following table shows the greatest value of p for which the tests considered are asymptotically unbiased if

$$\Delta(\Phi, F) = \inf_{\mu} \sup_x \left| \Phi(x) - F(\mu + \sigma x) \right| \geq A n^{-p}$$

with $0 < A < \infty$:

	Q_n -test	χ^2 -test	Kimball's test	Moments ($\mu'_3 \neq 0, \mu'_4 \neq 3$)
$F \in H$	$\frac{1}{4}$	$\frac{1}{2} - \delta$	$\frac{1}{4}$	$\frac{1}{2}$
$F \in L$	$\frac{1}{4}$	$\frac{1}{2} - \delta$	$\frac{1}{4}$	$\frac{1}{2}$

The same results hold for a number of "realistic" measures of the distance, Δ , between the null and alternative distributions. For example, the same result will hold if we put:

$$\Delta^2 = \inf_{\mu, \sigma} \int_{-\infty}^{\infty} \{F(\mu + \sigma x) - \Phi(x)\}^2 \Psi(x) dx$$

where $\Psi(\cdot) > 0$ is one of the large class of weight functions.

It can easily be shown that in the case of m'_3 and m'_4 we obtain the same result in the table above if the test statistics are changed to $S^{-3} m_3$ and $S^{-4} m_4$ respectively. The behaviour of Q_n is however totally different and this is no doubt due to the fact that Q_n and S are highly correlated. It appears from the work of Venter (1968) that the lowest value of p for which $n^p(Q-S)$ will have a non-degenerate distribution is $p = 1$ and, if this is the case, the test based on Q will have power for alternatives in H and L in $K(1/2)$.

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